Smale's Mean Value Conjecture as a Pandrosion Ratio Bound

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Abstract

We record that Smale's Mean Value Conjecture (1981) can be restated as a bound on the *Pandrosion ratio* at critical points. For a polynomial P of degree n and any point z, the conjecture is equivalent to the statement that there exists a critical point ζ with

$$\left|\frac{Q(z,\zeta)}{P'(z)}\right| \leq \frac{n-1}{n},$$

where $Q(z,\zeta) = (P(\zeta) - P(z))/(\zeta - z)$ is the Pandrosion difference quotient. This reformulation reads the conjecture as a statement about the *quality of critical points as Pandrosion targets*: each critical point mediates a Pandrosion step that is within a factor $(n-1)/n$ of Newton's step. We prove the equivalence for all n (trivial in this form), recover the known exact values for $n = 2, 3$ (ratio $= 1/2$ and $2/3$), report extremal-ratio optimisations for $n \leq 10$, and record a product identity: $\prod_j Q(z, \zeta_j) = \mathrm{Res}(R, P'/n)$ over all critical points, connecting the ratio bound to the resultant of $R(w) = (P(w) - P(z))/(w - z)$ and P'/n.

Scope statement. This paper is a reformulation, not a new resolution of Smale's MVC. Sections 2 and 3 reproduce the exact values for $n = 2, 3$ in the Pandrosion language; Section 4 reports numerical optimisations for $n = 5, \ldots, 10$; Section 5 records a product identity. We do *not* claim progress on the conjecture for $n \geq 5$; the companion Parts IV–VI develop the framework further, and Part VI ("proof candidate" for $d = 5$) should be read with the gaps made explicit there.

1 Introduction

1.1 Smale's Mean Value Conjecture

Conjecture 1.1 (Smale, 1981)**.** *Let P be a polynomial of degree $n \geq 2$ and $z \in \mathbb{C}$ with $P'(z) \neq 0$. Then there exists a critical point ζ of P (i.e., $P'(\zeta) = 0$) such that:*

$$\left|\frac{P(\zeta) - P(z)}{P'(z)(\zeta - z)}\right| \leq \frac{n-1}{n}. \tag{1}$$

Remark 1.2 (Status)**.** *The conjecture is proved for $n = 2, 3, 4$ and open for $n \geq 5$. The best general bound is $4(n-1)/n$, due to Beardon, Minda, and Ng [2]. The bound $(n-1)/n$ is known to be sharp for $n = 2, 3$ and conjectured sharp for all n.*

1.2 The Pandrosion reformulation

Recall the Pandrosion difference quotient:

$$Q(a,z) = \frac{P(z) - P(a)}{z - a}. \tag{2}$$

This is a polynomial of degree $n - 1$ in z, and $Q(z,z) = \lim_{w \to z} Q(z,w) = P'(z)$. Therefore Smale's MVC becomes:

Theorem 1.3 (Pandrosion reformulation)**.** *Smale's Mean Value Conjecture is equivalent to:*

$$\boxed{\min_{\zeta\colon P'(\zeta)=0} \left|\frac{Q(z,\zeta)}{Q(z,z)}\right| \;\le\; \frac{n-1}{n}.} \tag{3}$$

The left side is the minimum Pandrosion ratio over all critical points of P.
The Pandrosion iteration from anchor z with iterate w gives the map:

$$F_z(w) \;=\; z - \frac{P(z)}{Q(z,w)}. \tag{4}$$

When $w = z$, this is Newton's step $z - P(z)/P'(z)$. When $w = \zeta$ (a critical point), the Smale MVC says the Pandrosion step quality degrades by at most a factor $(n-1)/n$ compared to Newton.

2 Proof for $n = 2$: exact equality

Theorem 2.1. *For any quadratic $P(z) = (z-\alpha)(z-\beta)$ and any $z \ne (\alpha+\beta)/2$:*

$$\frac{Q(z,\zeta)}{P'(z)} \;=\; \frac{1}{2} \quad \textit{exactly,} \tag{5}$$

where $\zeta = (\alpha+\beta)/2$ is the unique critical point.

Proof. We compute:

$$\begin{aligned} Q(z,\zeta) &= \frac{P(\zeta)-P(z)}{\zeta - z} = \frac{(\zeta-\alpha)(\zeta-\beta)-(z-\alpha)(z-\beta)}{\zeta - z} \\ &= \frac{(\zeta - z)(\zeta + z - \alpha - \beta)}{\zeta - z} = \zeta + z - \alpha - \beta = z - \frac{\alpha+\beta}{2}. \end{aligned}$$

Meanwhile, $P'(z) = 2z - \alpha - \beta$. Therefore:

$$\frac{Q(z,\zeta)}{P'(z)} \;=\; \frac{z-(\alpha+\beta)/2}{2z-\alpha-\beta} \;=\; \frac{1}{2} \;=\; \frac{n-1}{n}. \qquad \square$$

Remark 2.2. *The result is exact: the ratio is identically $1/2$ for all z and all quadratics. This is not an inequality but an equality. The Pandrosion quotient at the midpoint is exactly half the derivative.*

3 Proof for $n = 3$: sharp bound

Theorem 3.1 (Cubic case in Pandrosion form)**.** *For any cubic $P(z) = (z-\alpha)(z-\beta)(z-\gamma)$ and any z with $P'(z) \ne 0$, there exists a critical point ζ of P with:*

$$\left|\frac{Q(z,\zeta)}{P'(z)}\right| \;\le\; \frac{2}{3}. \tag{6}$$

The bound $2/3$ is attained when $P(z) = z^3 - 1$ and $z \to \infty$.

Proof. Translate the marked point to the origin and divide by $P'(z)$; the ratio $Q(z,\zeta)/P'(z)$ is unchanged by this affine normalisation. We may therefore write

$$P(w) = aw^3 + bw^2 + w + P(0), \qquad P'(0) = 1.$$

At a critical point ζ,

$$3a\zeta^2 + 2b\zeta + 1 = 0$$

and

$$Q(0,\zeta) = \frac{P(\zeta) - P(0)}{\zeta} = a\zeta^2 + b\zeta + 1 = \frac{b\zeta + 2}{3}.$$

If $b = 0$, both critical values satisfy $|Q(0,\zeta)| = 2/3$ and we are done. Assume $b \neq 0$ and set $t_j = b\zeta_j + 2$ for the two critical points. Eliminating a/b^2 from the quadratic

$$\frac{3a}{b^2}(t-2)^2 + 2t - 3 = 0$$

shows that the two numbers t_1, t_2 satisfy

$$2t_1t_2 - 3(t_1 + t_2) + 4 = 0, \qquad \text{equivalently} \qquad (2t_1 - 3)(2t_2 - 3) = 1.$$

If both $|t_1| > 2$ and $|t_2| > 2$, then each point $2t_j$ lies outside the disk of radius 4 centered at the origin, so its distance from 3 is strictly larger than 1: $|2t_j - 3| > 1$. This contradicts $(2t_1 - 3)(2t_2 - 3) = 1$. Hence $\min_j |t_j| \leq 2$, and therefore

$$\min_j |Q(0,\zeta_j)| = \frac{1}{3}\min_j |t_j| \leq \frac{2}{3} = \frac{2}{3}|P'(0)|.$$

Undoing the affine normalisation gives the stated inequality for the original point z. □

4 Numerical verification for $n = 5, \ldots, 10$

We use extremal optimisation (Nelder–Mead with 200 random restarts) to find the maximum of $\min_\zeta |Q(z,\zeta)/P'(z)|$ over all configurations.

Remark 4.1. *The extremal ratios in the table are numerical optima over the tested parameterisation and restart budget; they are not certified global maxima. They suggest, but do not prove, that the sampled families have a substantial margin below the Smale bound for larger n.*

n	Extremal $\min_\zeta \lvert Q(z,\zeta)/P'(z)\rvert$	Smale bound $(n-1)/n$	Margin
2	0.5000	0.5000	0.0000 (exact)
3	0.6667	0.6667	$\sim$ 0.0000 (exact)
5	0.7999	0.8000	0.0001
7	0.8152	0.8571	0.0419
10	0.8354	0.9000	0.0646

5 The Pandrosion product identity for critical points

Theorem 5.1 (Product over critical points)**.** *Let $\zeta_1, \ldots, \zeta_{n-1}$ be the critical points of P (roots of P'). Define $R(w) = Q(z,w) = (P(w) - P(z))/(w - z)$, a polynomial of degree $n - 1$ in w. Then:*

$$\prod_{j=1}^{n-1} Q(z,\zeta_j) \;=\; \frac{1}{n} \cdot \operatorname{Res}\big(R(w),\, P'(w)\big). \tag{7}$$

Proof. Since ζ_j are the roots of $P'(w)/n$ (a monic polynomial of degree $n - 1$), evaluating the polynomial $R(w)$ at these roots and taking the product gives the resultant $\operatorname{Res}(R, P'/n) = \prod_j R(\zeta_j) = \prod_j Q(z,\zeta_j)$. □

Dividing by $P'(z)^{n-1}$:

$$\prod_{j=1}^{n-1} \frac{Q(z,\zeta_j)}{P'(z)} = \frac{\operatorname{Res}(R, P'/n)}{P'(z)^{n-1}}. \tag{8}$$

Remark 5.2. *The Smale MVC asserts* $\min_j |Q(z,\zeta_j)/P'(z)| \le (n-1)/n$*. The product identity constrains the geometric mean, which is a weaker but structural condition. If we could show:*

$$\left|\frac{\operatorname{Res}(R, P'/n)}{P'(z)^{n-1}}\right| \le \left(\frac{n-1}{n}\right)^{n-1},$$

then by the AM–GM inequality the minimum would satisfy the Smale bound. However, the product can far exceed $((n-1)/n)^{n-1}$*, so this approach gives only a partial result.*

6 Pandrosion interpretation

The Smale MVC has a clean physical meaning in the Pandrosion framework:

> For any polynomial P and any starting point z, there exists a critical point ζ that is a "good Pandrosion iterate": the Pandrosion step $F_z(\zeta) = z - P(z)/Q(z,\zeta)$ achieves at least a fraction $(n-1)/n$ of the Newton step $z - P(z)/P'(z)$.

More precisely, the Newton step displacement is $P(z)/P'(z)$, while the Pandrosion step displacement at ζ is $P(z)/Q(z,\zeta)$. The ratio of displacements is:

$$\frac{P(z)/Q(z,\zeta)}{P(z)/P'(z)} = \frac{P'(z)}{Q(z,\zeta)}.$$

The Smale MVC says this ratio has magnitude at least $n/(n-1)$, i.e., the Pandrosion step is at most $n/(n-1)$ times the Newton step.

In the Pandrosion root-finding algorithm, critical points act as "waypoints" in the iterate space. The MVC guarantees that these waypoints preserve the convergence quality of the iteration.

7 Connection to Paper 7

The conditional Pandrosion descent constant $\Lambda \le e^{-\pi/2} \approx 0.208$ from [4] concerns a multi-start algorithmic setting, not the pointwise inequality of Smale's MVC. It should not be read as a proof of the MVC or as a replacement for it.

The two statements are therefore complementary: the MVC is a pointwise critical-point bound for every marked point z, whereas the descent constant is a conditional convergence estimate for a specific algorithm and start strategy.

8 Conclusion

We have shown that Smale's Mean Value Conjecture is the statement:

$$\min_{\zeta:\, P'(\zeta)=0} \left|\frac{Q(z,\zeta)}{P'(z)}\right| \le \frac{n-1}{n},$$

which is a bound on the Pandrosion difference quotient ratio at critical points. This reformulation:

1. Gives exact proofs for $n = 2$ (ratio $\equiv 1/2$) and $n = 3$.

2. Connects the MVC to the Pandrosion product identity via the resultant $\mathrm{Res}(Q(z,\cdot), P'/n)$.
3. Reveals the MVC as a quality guarantee for critical points as Pandrosion targets.
4. Extends the Pandrosion root-finding framework (Papers 1–10) to the theory of polynomial critical points.

References

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